

I M Smrti — How the Data Works, and Where It Breaks

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1. What the product is

I M Smrti (imsmrti.app) — a health-record app for the person in an Indian family who manages everyone else's health. The son or daughter chasing prescriptions and lab reports for ageing parents across three different doctors.

You add a family member, photograph their reports, and the app:

- stores everything securely (12 Indian languages, DPDP-compliant)
- explains each report in plain spoken Hindi/Tamil/etc. — not textbook language
- generates a one-page briefing to hand the doctor before an appointment
- shows an emergency QR card (blood group, allergies, medicines) anyone can scan
- **turns reports into structured data, charts trends, and tells you where your

health is heading** ← the part this document is about

Stack: React + Firebase (Firestore) + Google Gemini. The app is live and working; the open problems are almost entirely data problems, not app problems.

2. How data actually flows

```
photo/PDF upload
  ↓
Gemini (one call) → returns BOTH:
  |               • a plain-language summary for the human
  |               • structured JSON: metrics / medicines / follow-ups
  ↓
validation guard → rejects anything that isn't a real medical measurement
  ↓
Firestore → health_metrics · medications · follow_ups
  ↓
  ↳ trend charts (per test, per patient)
  ↳ health trajectory engine (score, direction, suggested actions)
  ↳ daily background agent (abnormal results, overdue follow-ups, refills)
```

The collections

`health_metrics` — one row per lab value

```
{
  "userId":"...", "patientId":"...",          // account + WHICH family member
  "documentId":"...", "documentName":"Ritu test 2026.pdf",
  "test":"Vitamin D",                          // canonical name – groups the trend
  "testRaw":"VITAMIN D (25-OH)",               // exactly as printed
  "value":8.4, "unit":"ng/ml",
  "refLow":30, "refHigh":100,
  "status":"low",                             // low | normal | high | unknown
  "date":"2026-02-05"
}
```

`medications` — { name, dose, frequency, durationDays, startDate, expectedEndDate } `follow_ups` — { advice, dueDate, status } (drives "your review is 12 days overdue") `patients` — { name, gender, dob, bloodGroup, allergies, conditions, relationship }

Every record carries both `userId` and `patientId`. Mixing patients would be a serious bug — a father's cholesterol must never appear on his daughter's chart.

3. The validation guard (and why it exists)

A business PDF — a client pricing plan — was once mined for numbers, and **₹60,000 / ₹1,20,000 / ₹3,00,000 were saved as "Weight" readings in INR** and charted. Separately, a Vitamin D result was stored with a reference range of `null-1` when the real range is 30–100 ng/mL.

That's the kind of failure that makes someone stop trusting a health app entirely, so there's now a guard that every extracted number must pass:

- **Non-clinical documents produce zero metrics.** No exceptions.
- **Currency, phone numbers, IDs, durations, counts** are rejected outright.
- **Physiological plausibility** — a 450 kg body weight or a 9-digit "value" is

thrown out.

- **Bogus reference ranges get repaired** from a trusted table rather than trusted

blindly from the model.

It's verified against 12 cases including the exact garbage that reached production. It passes all 12.

Important limitation: a guard only *blocks bad data*. It does nothing to *produce good data*. That's problem #1 below.

4. The genuinely hard problems (the interesting bit)

Problem 1 — Coverage, not accuracy

Of 48 documents: **26 were never AI-analysed at all**, 12 are structured, and 10 have summaries that haven't been converted to structured data. I originally assumed extraction was broken; the real gap is that most documents never went through the pipeline. Backfill now runs in batches of 40 and continues until done — but the 26 unanalysed ones need fresh AI calls against the original images, which costs money per document.

Open question: what's the right re-processing strategy when a user arrives with 200 old reports? Batch overnight? Charge for it? Process only what's recent?

Problem 2 — No canonical medical vocabulary

The same test is printed differently by every lab, so trend lines silently split into separate series instead of forming one:

```
"HBA1C" · "HbA1c" · "Glycated Haemoglobin" · "A1c"           → must be ONE concept
"VITAMIN D (25-OH)" · "Vit D3" · "25-Hydroxyvitamin D"       → ONE concept
```

I've hand-built **66 tests** (attached: `lab_vocabulary_current.csv`) covering CBC, lipids, LFT, KFT, thyroid, sugar, vitamins, electrolytes, urine routine. Each has: canonical name, accepted units, normal range, plausibility bounds, and a plain-language explanation.

Two things I know are weak:

1. **Ranges are adult-generic.** They don't vary by age or sex, and they genuinely should (Hemoglobin, Ferritin, Creatinine, HDL all differ by sex; children differ a lot). In a health app a wrong "normal" is dangerous — it can make a serious value look fine.

1. **They're my best knowledge, not clinically reviewed.** This needs a doctor or pathologist to sign off, not an engineer.

Problem 3 — Drug names are a mess

Real strings sitting in the database right now:

Extracted as	Probably is	Issue
Vitad3 60K	Cholecalciferol 60,000 IU	dose buried in brand name
Mouture LC	Montair/Montek LC (Montelukast + Levocetirizine)	OCR error
Grahaconeazole	likely Itraconazole	OCR error, unusable
Amrol Fin, Zine 200 mg, Contimega 0 Lo	?	unresolved brands

dose and frequency come back empty most of the time. Mapping Indian brand names → generic molecule + strength is a real data problem I haven't solved.

Problem 4 — Accuracy is unmeasured

There is **no ground truth set**, so nobody can say whether extraction is 40% accurate or 90%. That's precisely how the rupee-amount bug survived. The fix is unglamorous: 30–50 real reports, hand-labelled, used as a ruler. Without it, every "improvement" is a guess.

5. What's already built on top

- **Trend charts** per test per patient, with reference bands and source document
- **Health trajectory engine** — computes a 0–100 attention score, whether each

value is moving toward or away from the healthy range, and suggests concrete actions grouped by what shares a response (sugar / cholesterol / vitamins / anaemia / kidney / liver / thyroid / BP). Purely computed from the person's own records — no AI guessing on the numbers.

- **Daily background agent** — flags abnormal results, worsening 3-reading trends, overdue doctor follow-ups, and medicines about to run out.

6. Where you'd fit, when the time comes

If this reaches real volume — thousands of documents a month — the data layer becomes a genuine job:

1. **Medical vocabulary at scale** — LOINC mapping, drug normalisation, age/sex-aware ranges. The highest-value piece.

1. **Accuracy measurement + monitoring** — gold-standard sets, precision/recall per test, dashboards showing which labs and formats fail.

1. **De-identification + aggregation** — this is the ambition I'm most excited about:

understanding patterns in Indian health at population scale (what conditions by age group, which medicines, which deficiencies) and eventually sharing that with researchers or public-health bodies. Done properly it means k-anonymity, minimum group sizes, banded ages, no free text — real privacy engineering.

On that last one, the legal reality: our privacy policy says data is "*strictly used to provide medical record storage... generate AI health insights.*" Research is a **different purpose** under DPDP 2023 and needs **separate opt-in consent** — which we're adding, off by default. And "without a name" is not anonymous: age + city + a rare condition re-identifies people. So this is a someday-at-proper-scale thing, built correctly, not a shortcut.

7. What would actually help right now (no obligation)

Nothing paid, nothing time-consuming — just if any of it interests you:

1. **Poke holes in the data model.** Anything in section 2 you'd design differently?
2. **Look at the vocabulary CSV** — especially whether the age/sex problem is as

significant as I think, and how you'd structure that.

1. **Tell me if problem 4 (measuring accuracy) is where you'd start too**, or if

you'd attack something else first.

Honest answers are worth more to me than polite ones. If you think the whole approach is wrong, that's the most useful thing you could tell me.

Attached: `lab_vocabulary_current.csv` — the 66 tests, ranges and plain-language explanations currently powering the app.